IILM UNIVERSITY
GREATER NOIDA, UTTAR PRADESH

AICTE – EDUSKILLS VIRTUAL INTERNSHIP

Agribusiness Performance Analytics

AgriPulse – 4-Week Professional Internship Presentation Report

INTERNSHIP DURATION

29 June 2026 – 26 July 2026 (4 Weeks)

STUDENT CREDENTIALS

Ujjwal Kumar

ROLL NO.	PROGRAM	SECTION.
25SCS1003001220	2CSE1B Tech (CSE)	

AGRIBUSINESS PERFORMANCE ANALYTICS

AgriPulse

Transforming Agricultural Raw Data into Actionable Business Intelligence

Project Type: Data Analytics / Agribusiness Analytics | Role: Junior Data Analyst | Duration: 4 Weeks

AGRIPULSE ANALYTICS

Project Overview & Key Objectives

Executive Summary

AgriPulse is a comprehensive 4-week Agribusiness Data Analytics internship project demonstrating the power of data-driven techniques in modern agriculture.

By capturing end-to-end operational variables, the project focuses on boosting yield productivity, cost efficiency, supply-chain responsiveness, and strategic decision-making.

- **Yield & Productivity**
  Monitor crop yield per hectare and total output across diverse plots.
- **Cost Efficiency**
  Evaluate production expenses, cost per unit, and gross profit margins.
- **Market Dynamics**
  Understand seasonal price trends and volatility for crop planning.
- **Weather Impacts**
  Quantify climate, rainfall, and temperature influences on harvest output.
- **Supply-Chain Health**
  Pinpoint storage bottlenecks, inventory turnover, and post-harvest wastage.

AGRIPULSE ANALYTICS

4-Week Agribusiness Analytics Lifecycle

WEEK 1

Data Strategy & Planning

- Defined business objectives and core domain KPIs
- Identified primary & secondary agricultural data sources
- Established data governance & implementation roadmap

WEEK 2

Acquisition & Preprocessing

- Structured multi-source data ingestion pipeline
- Handled missing values and eliminated duplicate records
- Normalized metrics & performed Z-score outlier detection

WEEK 3

Analysis & Visualization

- Exploratory & statistical analysis on crop performance
- Yield vs. cost correlation & market price trend modeling
- Designed static charts and interactive dashboards

WEEK 4

Evaluation & Improvement

- KPI tracking & performance benchmark comparison
- Formulated action items for supply chain & crop selection
- Defined automated alerts & long-term platform roadmap

Data Engineering & Technologies

Data Preprocessing Workflow

1. Data Ingestion
Collected disparate datasets across crop yields, market rates, weather logs, and logistics.

2. Cleaning & Imputation
Resolved missing data using median values and removed identical duplicate entries.

3. Validation & Quality Assurance
Verified logical range bounds (e.g., non-negative yields, valid pH values).

4. Normalization & Scaling
Applied MinMax and Standard scaling for consistent statistical comparison.

5. Outlier Detection
Leveraged Z-Score and Interquartile Range (IQR) filtering to handle anomalies.

Technologies & Tools

- **Python**
  Core language for data processing
- **Pandas & NumPy**
  Data manipulation & numerical computation
- **Matplotlib & Seaborn**
  Static data visualization & plotting
- **Streamlit**
  Interactive web dashboard interface
- **Git & GitHub**
  Version control & portfolio hosting
- **Excel / Word**
  Documentation & baseline reporting

AGRIPULSE ANALYTICS

Key Performance Indicators (KPIs)

Production & Yield Metrics

- **Yield per Hectare:** Measures land productivity and soil health
- **Production Quantity:** Total harvest output in metric tons

Financial & Cost Metrics

- **Production Cost:** Total input costs (fertilizers, labor, seeds)
- **Cost per Unit:** Unit economics per ton produced
- **Revenue & Gross Margin:** Profitability across crop types

Supply Chain & Operational Metrics

- **Inventory Turnover:** Efficiency of stock movement
- **Wastage Rate:** Post-harvest loss percentage
- **Delivery Performance:** On-time distribution fulfillment

Market & Climate Metrics

- **Market Price Variation:** Track wholesale price shifts
- **Weather Impact Score:** Correlation with rainfall & temperature

AGRIPULSE ANALYTICS

Key Analytical Insights & Findings

INPUT OPTIMIZATION

Cost vs. Yield Trade-offs

Statistical profiling demonstrated that increasing input expenditure beyond optimal thresholds yields diminishing returns in crop output.

CLIMATE RESILIENCE

Weather Vulnerability Mapping

Correlation analysis highlighted significant yield variances during unseasonal temperature spikes, emphasizing irrigation needs.

SUPPLY CHAIN EFFICIENCY

Wastage Reduction Targets

Inventory data revealed that delivery delays over 48 hours increased post-harvest wastage rates by over 15%.

STRATEGIC TIMING

Market Price Dynamics

Historical price variations indicated predictable high-margin seasonal windows for selling key cash crops.

Process Recommendations & Future Scope

Actionable Recommendations

✓ Crop & Input Planning
Reallocate inputs toward high-margin crops and adjust localized fertilizer usage.

✓ Inventory Control
Adopt Just-In-Time storage protocols to reduce transit delay and post-harvest decay.

✓ Market Synchronization
Align harvest timelines with predicted peak-market price windows.

✓ Data Quality Protocols
Standardize field-level data collection for higher analytical reliability.

Future Scope & Platform Growth

- 🚀 Machine Learning yield prediction & crop demand forecasting
- 🚀 Weather-risk analysis & automated price prediction models
- 🚀 Satellite and remote-sensing IoT data integration
- 🚀 Automated real-time KPI alerts & logistics route optimization
- 🚀 Full-scale interactive cloud dashboards (Streamlit / Power BI)

PROJECT OUTCOME

Transforming Agribusiness with Data Insights

AgriPulse establishes a solid, scalable framework for turning complex agricultural datasets into actionable business intelligence. Thank you for exploring this project repository!